Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 1 Quiz 1 - Kinematics in One Dimension

Mock Exam B · Form 1: Single-step application

Multiple Choice Questions: Select the best answer. (1 point each)

1. Two position graphs cross. A slopes up at the crossing; B slopes down. What must be true at the crossing?

(A) Equal speeds.

(B) Both stop.

(C) Same position, opposite velocity directions.

(D) Equal accelerations.

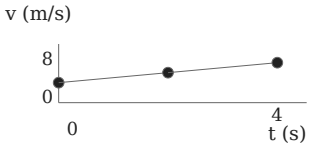
2. A velocity–time graph is a straight line from +4 m/s at t = 0 to +8 m/s at t = 4 s. What is the displacement?

(A) 12 m

(B) 24 m

(C) -24 m

(D) 48 m



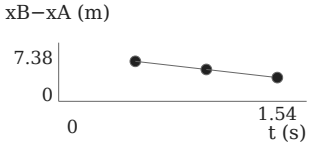
3. A runs at +5 m/s and B at +2 m/s. Record the lead of B ahead of A as a signed separation. t (s) -> xB-xA (m): (0.54, 7.38); (1.04, 5.88); (1.54, 4.38). What signed vertical change follows a decrease of 0.25 horizontally?

(A) 0.75

(B) 1.125

(C) 0.375

(D) 1.5



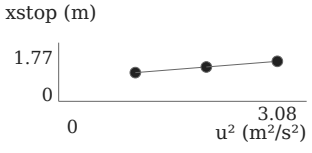
4. A cart brakes immediately at a constant magnitude of 2 m/s². Read its stopping coordinate along the track. u² (m²/s²) -> xstop (m): (1.08, 1.27); (2.08, 1.52); (3.08, 1.77). Predict the reading at horizontal coordinate 2.58.

(A) 0.987

(B) 1.645

(C) 2.961

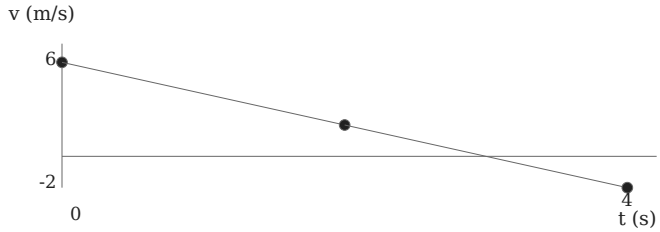
(D) 2.303



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A sensor records a cart with x0 = -3 m and v0 = +6 m/s. It has constant a = -2 m/s² for 4 s. Right is positive.

a. Describe the shapes and initial values of x–t, v–t, and a–t. **(1 pt)**



b. Find final position, displacement, and distance; explain any reversal. **(3 pt)**

6. A bicycle moving right brakes uniformly. Use the model only until it stops. t (s) -> v (m/s): (0.54, 8.92); (1.04, 7.92); (1.54, 6.92).

a. Build the model for brake recorder from physics and predict both slope and intercept. **(2 pt)**

b. Label the axes and units and demonstrate interpolation between two readings. **(2 pt)**

7. A cart’s velocity changes from u to v under constant acceleration over time t. Use the v–t graph to derive average velocity and displacement. **(3 pt)**